

Wasi Omar Syed

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Education

Islamic University of Technology (IUT)

Boardbazar, Gazipur, Bangladesh

B.Sc. in Software Engineering

Sep 2024 – Sep 2028 (Expected)

CGPA: 4.00 / 4.00 (through 4th semester, currently in 2nd year)

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Structured Programming, Database Management Systems I & II, Discrete Mathematics, Linear Algebra, Differential Calculus, Probability & Statistical Inference

Research Experience

Student Researcher (Self-Initiated, Student-Led Group)

Ongoing (2026)

Spatio-Temporal Event Labeling for LLM Reasoning

Islamic University of Technology

- Building an annotation tool for labeling spatio-temporal events and their relative timing in narrative text, intended to train language models to reason about temporal event structure
- Applying Allen's Interval Algebra as the theoretical foundation for representing temporal relations between annotated event spans
- Working independently within a small student-led research group (under senior students); project is in active development with no publication output yet

Publications

No publications at this time. A working paper on spatio-temporal event labeling is in early-stage development (see Research Experience).

Research & Technical Projects

CareerPilot — AI-Powered Job Search & Career Guidance Platform

May 2026

Team Lead, 3-member team

GitHub | Live Demo

- Led full project ideation, system architecture, and technical direction across all subsystems, from web crawling pipelines to AI-driven search
- Directed integration of an AI search layer for job matching and discovery, coordinating design decisions across the team
- Achieved 4th place at IUTCS CodeSprint 2026 (IUT Computer Society, May 12, 2026)

Driest_Run — Multi-Layered Safety Gate for Agentic Hardware Control

2026

Safety & Agentic Control Lead

GitHub | Live Demo

- Designed and implemented a multi-layered safety gate architecture to constrain agentic AI actions before hardware execution
- Built voice-based control and an agentic AI control pipeline to safely translate natural-language commands into physical hardware movement
- Placed 12th among 250+ teams at Techathon Nationals & Rover Summit, a hackathon powered by Programming Hero

PawFect Match — AI-Powered Pet Adoption Platform

2025

Project Lead

Live Demo

- Oversaw system design and implementation for a full-stack pet adoption platform with an AI-based owner-pet matching system
- Implemented a swipe-based (Tinder-style) adoption flow and real-time chat via WebSockets
- Runner-up, IUTCS CodeSprint 2025 Hackathon

Student-Driven Marketplace — Full-Stack AI-Powered Marketplace Platform

2025

Live Demo

- Built a full-stack marketplace enabling students to buy/sell items, with authentication, listings, messaging, and admin moderation
- Implemented JWT authentication, role-based access control, and REST APIs (Node.js, Express.js, MongoDB); real-time messaging via Socket.IO
- Integrated an AI assistant and smart search to improve user discovery; deployed on Vercel with a CI-style workflow

FeatherDB — Custom Relational Database Engine (C++17)

2025

Parser & Query Engine Lead

GitHub

- Designed and implemented a full SQL parsing pipeline from scratch (tokenizer → AST → query executor) with no external libraries
- Built a recursive query executor supporting nested subqueries in both **FROM** and **WHERE** clauses
- Enforced relational integrity at execution time via **PRIMARY KEY**, **UNIQUE**, and **FOREIGN KEY** constraints with a **RESTRICT** delete policy; implemented in-memory **ORDER BY** sorting via quicksort

Deep Learning Lab — CNNs, Dense Networks & Transformers

2025

GitHub

- Implemented a Transformer from scratch (encoder, decoder, positional embeddings) and applied it to an NLP sentiment analysis task (IMDB)
- Built and trained CNNs for image classification on CIFAR-10; compared dense networks vs. CNNs for image recognition performance

Hands-on Machine Learning — From-Scratch ML Algorithm Implementations

2025

GitHub

- Implemented core ML algorithms from scratch in Python/NumPy (linear/logistic regression, gradient descent, PCA, decision trees)
- Built full ML pipelines (EDA, preprocessing, training, evaluation) and compared classical models (decision trees vs. random forests, lasso vs. ridge) on real-world datasets

Technical Skills

- **Programming Languages:** Python, C++, Java, JavaScript
- **Machine Learning & AI:** PyTorch, Scikit-learn, YOLO, LangChain, LangGraph, CrewAI, FastMCP
- **Data Science & Visualization:** NumPy, Pandas, Matplotlib, Jupyter Notebook
- **Backend & Deployment:** FastAPI, Node.js, Express.js, Next.js
- **Frontend & App Development:** React, Next.js, Tailwind CSS, Flutter
- **Databases:** MongoDB, PostgreSQL
- **DevOps & Cloud:** AWS (basic services), Docker, Kubernetes (fundamentals), Terraform (IaC)
- **Tools & Platforms:** Git, GitHub, n8n
- **Mathematical & Statistical Foundations:** Probability, Bayesian Inference, Hypothesis Testing, Markov Models, Statistical Inference

Awards & Achievements

- **4th Place**, IUTCS CodeSprint 2026 Hackathon — *CareerPilot* (Team Lead) May 2026
- **12th of 250+ Teams**, Techathon Nationals & Rover Summit (powered by Programming Hero) — *Driest_Run* (Safety & Agentic Control Lead) 2026
- **Runner-up**, IUTCS CodeSprint 2025 Hackathon — *PawFect Match* (Project Lead) Jul 2025

Certifications

- **CS50's Introduction to Artificial Intelligence with Python**, HarvardX — completed 12 AI projects covering search algorithms, knowledge representation, optimization, machine learning, neural networks, and NLP Jul 2025
- **Data Structures & Algorithms in C++**, Apna College — problem solving and competitive programming 2025
- **Algorithms, Data Structures & Problem Solving**, YouKnowWho Academy 2025

References

Available upon request.